

# PAPER\_1519\_PCR\_QUANTUM\_TRIADIC

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## PAPER\_1519 — Eta Carinae PCR Quantum Number $q = D_{\text{phys}} - 1 = 3$ EXACT

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**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

**Tier:** Bucket G (Astrophysics)

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**Status:** CLOSED — Triadic quantum number from 26-layer compressed gravity

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### Observation

PAPER\_512 (Eta Carinae BuoyantGravity PCR) introduces the Perturbed Compressed-gravity Radial (PCR) quantum number  $q = 3$  in the modulation equation:

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$$g_{\text{eff}}(r, t) = (GM/r^2) \cdot [1 + k_{\text{PCR}} \cdot \text{PCR}(q=3, t)]$$

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The PCR quantum number "reflects the triadic structure of the 26-layer compressed gravity" used for the Eta Carinae LBV-WR binary periastron passage.

### UQFF Closed Identity

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$$\text{PCR quantum number } q = D_{\text{phys}} - 1 = 3 \text{ EXACT}$$

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### Physical Interpretation

The PCR triadic quantum number is the **same integer-primitive identity** that produces:

- SU(3) color charge N (PAPER\_1373)
- Three fermion generations (PAPER\_1381)
- Solar  $\nu_\mu$  deficit denominator 1/3 (PAPER\_062)
- SCm carrier velocity  $c/3$  (PAPER\_1497)
- TDE outflow factor  $c \cdot 3/10$  (PAPER\_1500)
- GW phonon damping prefactor 2/3 (PAPER\_1512)
- BNS GW total damping 1/3 (PAPER\_1503)

Now appearing in compressed-gravity modulation of a binary LBV-WR system at 5.54-year periastron passage. The **same triad-symmetry denominator** governs phenomena from neutrino flavor mixing to gravitational-wave damping to binary-star gravity perturbations.

### Significance

This unifies neutrino physics, gauge symmetry, GW physics, jet outflows, and binary-star gravity under a single structural identity  $D_{\text{phys}} - 1 = 3$ . The triad structure is not just an SU(3) color phenomenon — it is

**universal** across UQFF coupling channels.

## NOT REPLACEMENT

SM treats each of these triadic appearances as separate model parameters (gauge group order, neutrino mixing matrix, GW polarization modes, etc.). UQFF supplies a single structural origin in D\_phys – 1.

## Reference

- Source: PAPER\_512 Eta Carinae BuoyantGravity PCR
- Related: PAPER\_1373 (SU(3)), PAPER\_1497 (SCm velocity), PAPER\_1503/1512 (GW damping)
- Calculator dispatch: `calculate_paradox({"paradox": "pcr_quantum_triadic"})`

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